

Chapter 1

Introduction to IS

(Information System)

Basic IS Concepts

What is an IS ?

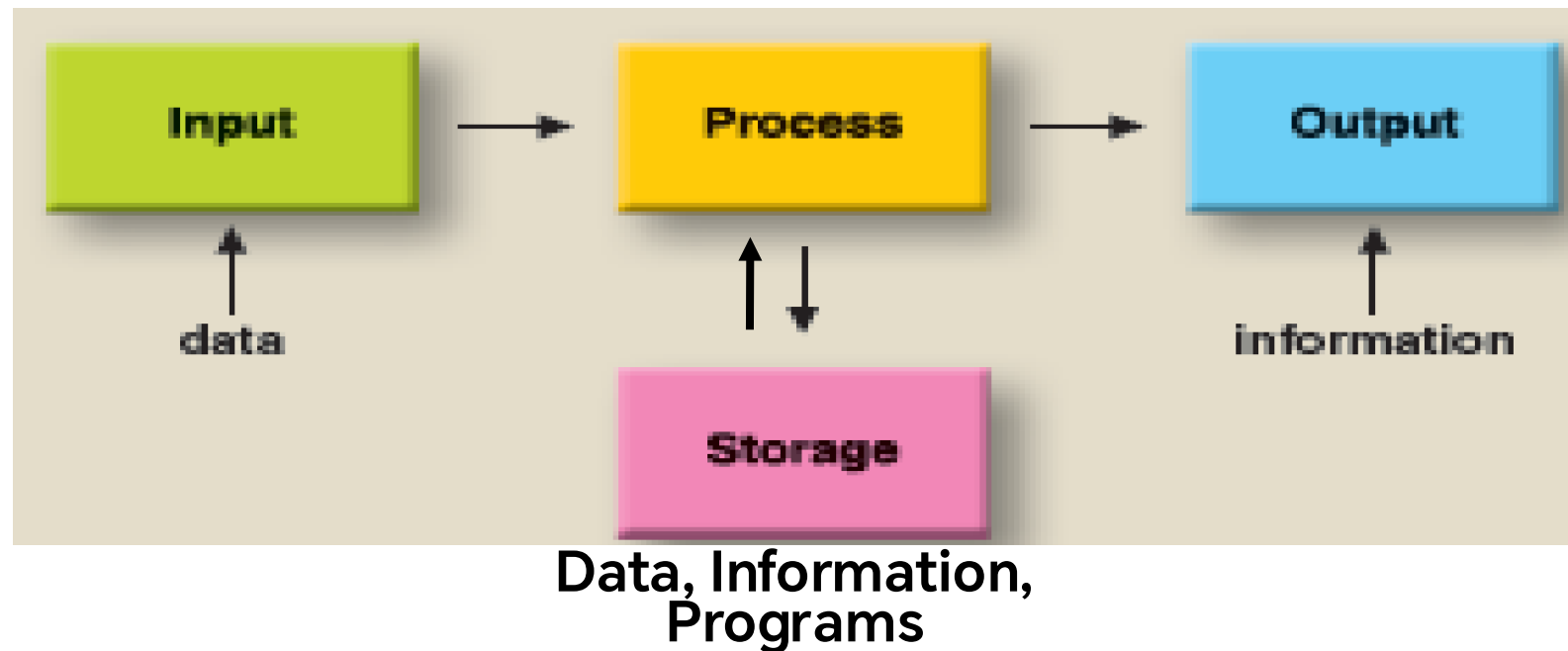
- **Components:**

- **Hardware** : servers, clients, network infrastructure
- **Software** : operating systems (os), DBMS and applications
- **Process** : consists of **activities** or tasks supported by the applications eg. create PO, make payments, receive goods
- **Data stored (database)** : files or tables (transactions or masters)
- **People** : end-users, IT/IS personnels

- **Purpose :**

- To capture, store, input, process data and to produce *information* to help operate, support and manage organization.

Functions of Information System



- 4 Functions performed by IS (above)
 - **Input** : accepts raw **data** from outside the system
 - **Process** : processes the raw data into information
 - **Output** : produces the results of the process (**information**)
 - **Storage** : stores *data, information, programs*

Example 1 – Inventory Control System (Receiving Goods Module)

- **Input function** – receive/accept Order and delivery (supplier) details (eg. Purchase Order No, Delivery Order no, item code, quantities) **product A 20K, 10K+10K**
- **Process function** – process the input data (ie. **goods received data**) and update the quantities in the inventory data store etc. **product A 10+10K=10010**
- **Output function** – the user (eg. store manager) retrieve information about the inventory details.
- **Storage function** – data (eg. goods received data) and information (updated inventory balance) stored for future use.

5 Components of an Information System

1. Hardware
2. Software
3. Processes
4. Data Stored
5. People

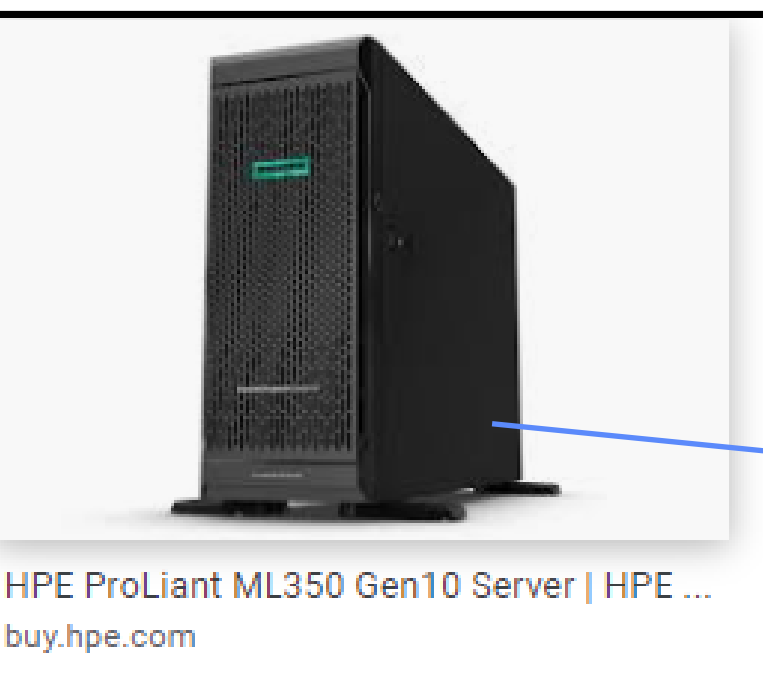
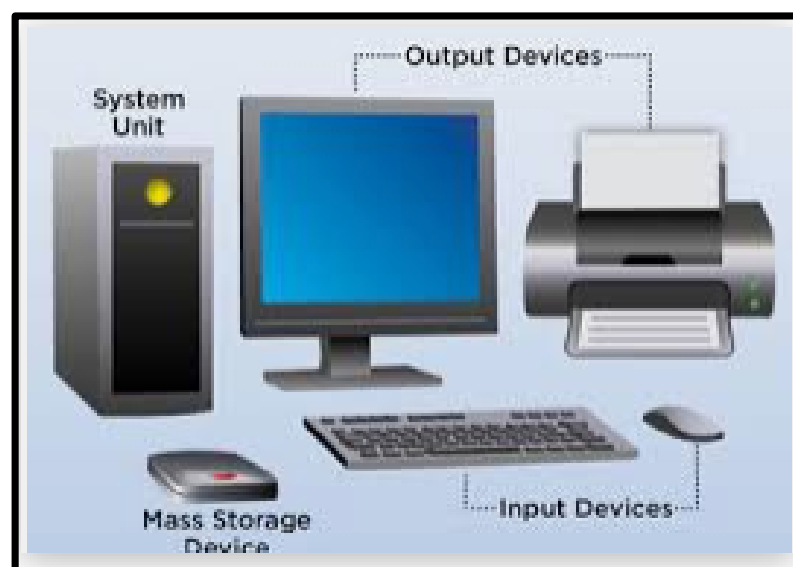
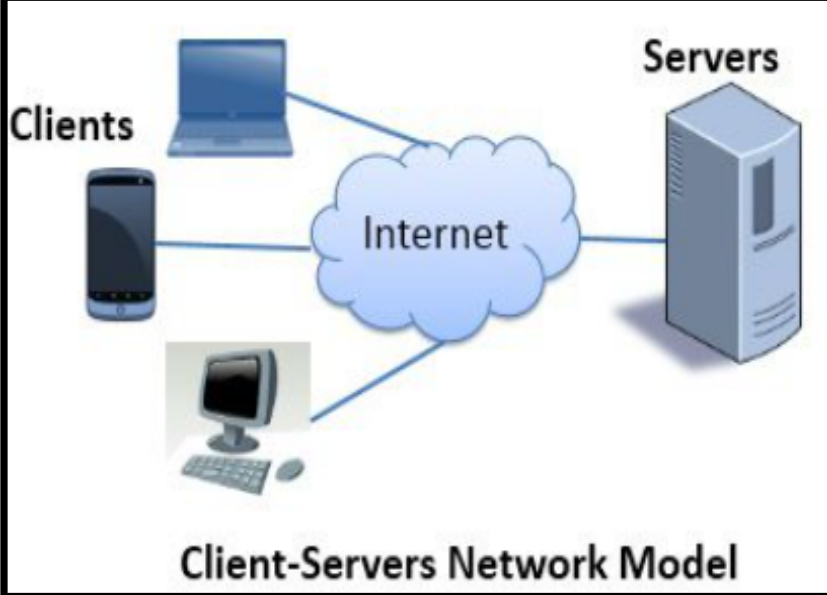


1. Hardware

- **Servers** – (eg. Mini-computers, mainframes)
- **Clients** – (eg. PC, smartphone, laptops, ATM etc)
- **Network Infrastructure** – (eg. wire, modems, hubs, routers, switches etc).
- **Peripherals** – (input, output, storage devices) eg. digital cameras, microphones, printers, readers, sensors, scanners etc

- A **server** is a host that provides information or a service to other devices on the network
- A computer or other device that receives a service is called a **client**
- In a **client-server network**, clients access information provided by one or more **servers**





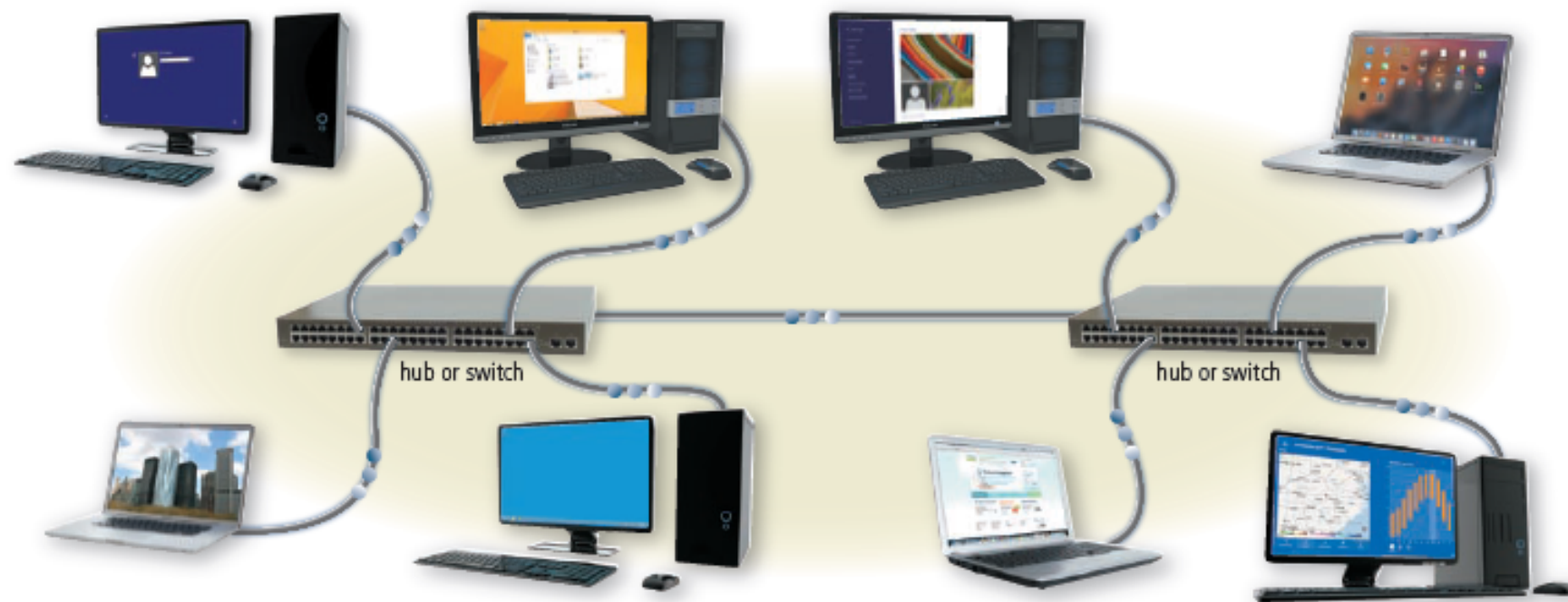
server

- **Network Infrastructure** – (eg. wire, modems, hubs, routers, switches etc).

A hub/ switch is a device that provides a central point for cables in a network.

Larger networks typically use a hub, while smaller networks use a switch.

Some hubs/switches include routers. That is, the hub/ switch receives data from many directions and then forward it to one or more destinations.



A router connects multiple computers or other routers together and transmits data to its correct destination on a network.

Example:

Home network – router allows multiple computers to share a single high speed Internet connection such as through a cable modem/DSL modem.

To prevent unauthorized users from accessing files and computer, many are protected by a build-in firewall.

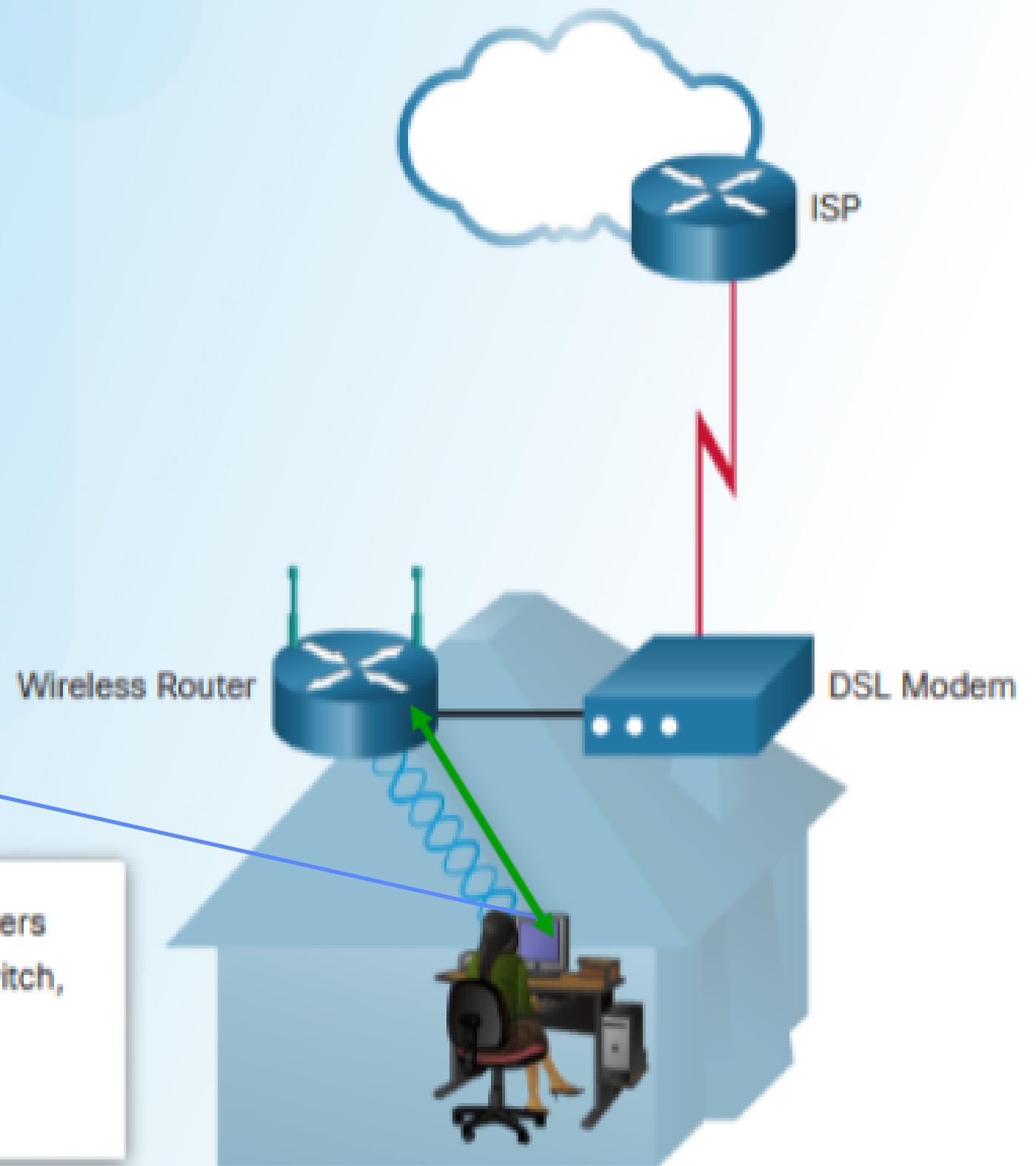


Typical Home Network

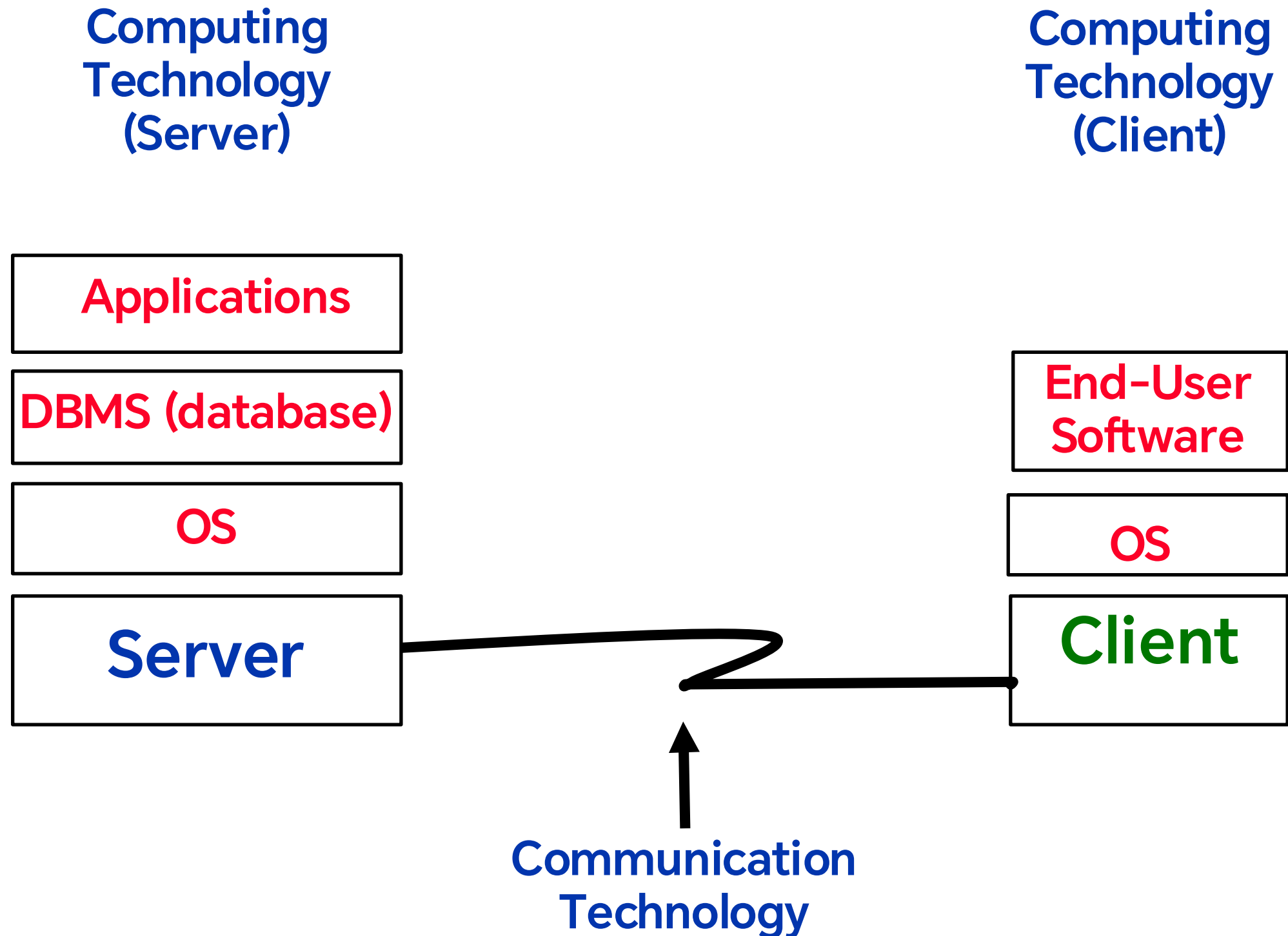
network interface cards (NICs)



In small businesses and homes, wireless routers perform the role of access point, Ethernet switch, and router.



ICT Technologies (Hardware and Software)



2. Software/program/app/software application

- **Operating System (OS)** – manages both the computer hardware and software resources of a computer system eg. Linux, Window, Unix, OS/400, Network OS.
- **Database Management System (DBMS)** – manage data stored. eg Oracle, DB2, SyBase.
- **Software Development Tools** – support programmers in developing programs eg. translators (eg compilers), tools (eg. CASE tools) and programming languages (eg. Python, C)
- **Application Systems** – supports the running of the various business processes eg. sales, purchasing, warehousing, manufacturing, HR systems.

Food point of sale program/ food order program processes and records the customer food order.

15

REG # 1 RON W. -- Plexis Point Of Sale

SALE TOTAL
20.58

CUST OPS - F8

TAKE OUT

F-11 F-12 F2 - ITEMS

DESCRIPTION QTY TOTAL T

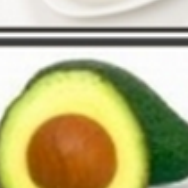
CHICKEN CLUB SANDWICH	1	5.99	T
PIZZA SLICE	1	1.99	T
TURKEY BURGER	1	4.99	T
TURKEY CLUB SANDWICH	1	5.99	T
MUSTARD	1	0.25	
MAYO	1	0.00	

GFT HLD D-IN

DISC CPN TCKTS

TIPS L lotto DWR

MAIN MENU

LUNCH MENU	NO PICTURES SAMPLE	DINNER MENU	BREAKFAST MENU	DRINKS	MUSTARD KETCHUP MAYO PICKLES ETC
					
					
					
					
TAXABLE ITEM	NON TAXABLE ITEM				

F1 - PAD F3 - CASH F4 - CREDIT F5 - PAY F6 - CASE/PK F7 ACTIONS F10 OPTIONS

NON-TAX DISCOUNT TTL TAX SUBTOTAL

MARINA BOOKSIDE CAFE 5:10:42 PM

3. Processes

- **Processes** – contain *steps* that tell people *how to use and operate* the system to perform tasks/activities involving *data*. **Process** : consists of *activities* or tasks supported by the applications eg. create PO, make payments, receive goods
- **Example** – Inventory Control System needs procedures :
 - How to enter **input** data
 - How to **process** the data into information
 - How to **output** information from the system.
 - How to **operate** the hardware and software.

4. Data Stored

- Data that is **processed** by the software (eg payroll application software/program), **retained** in the hardware (eg harddisk) and **used** by the people (eg end-users).
- **Example** – Inventory Control System, the stored data :
 - **Master File** (Stock File) eg Stock/product/item code, name, unit of measurement, quantity, unit price.
 - **Transaction File** (Goods Received File) eg. Goods Received number, date received etc.

1
8 **Data Stored /database** - It consists of all the data that is kept in computers in the system and that is used by the software of the system. Database Management System (DBMS) - manage data stored. eg Oracle, DB2, SyBase.



Database

Index	Address	City	State	Country	Phone
1	1000 Main Street	1000 Main Street	1000 Main Street	1000 Main Street	1000 Main Street
2	1000 Main Street	1000 Main Street	1000 Main Street	1000 Main Street	1000 Main Street
3	1000 Main Street	1000 Main Street	1000 Main Street	1000 Main Street	1000 Main Street
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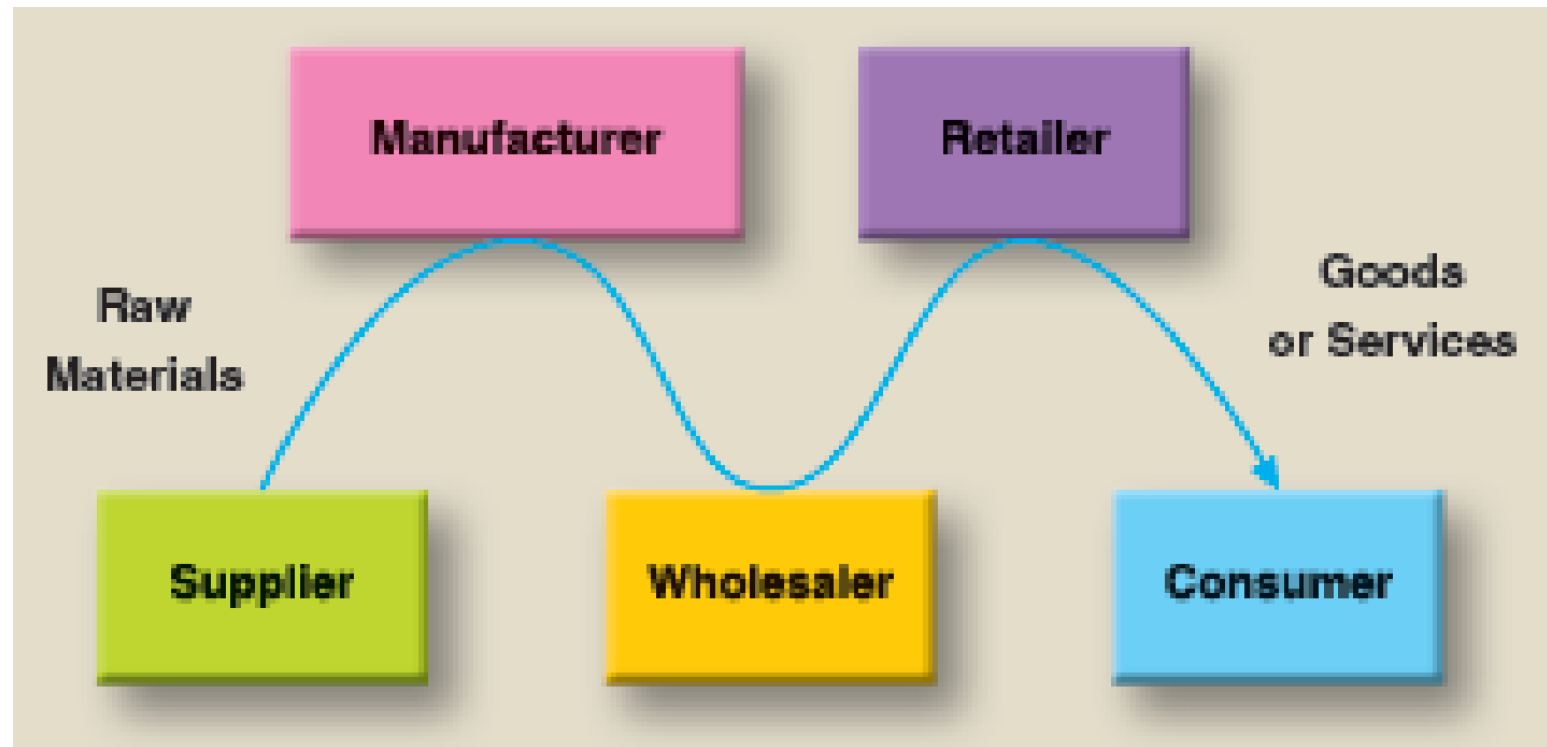
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5. People

- People are needed to run system.
 - *input* data (**Users**)
 - *receive* output information (**Management**)
 - *operate* the hardware and software (**Operation staff**)
- Example - Inventory Control System
 - **Users and Management** eg. *salesperson* (check stock), *purchasers* (check stock), *storehands* (receive and distribute goods), *store manager* (create inventory reports).
 - **Operation staff** – personnels operate the hardware and software in the system

Types of Organisations (Product)

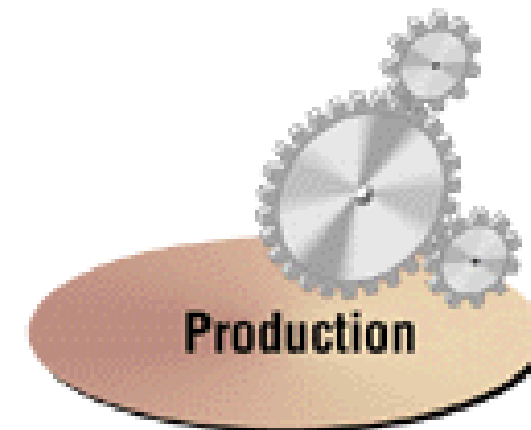
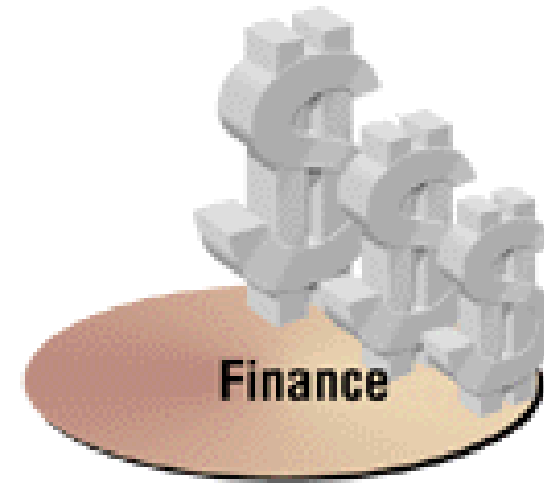


- **Supplier** – Supply raw materials that flow through an organisation and finally to consumer.
- **Manufacturer** – Raw materials, eg. oil and iron, are manufactured into products such as gasoline and automobiles.
- **Wholesaler** – These products are sold to other businesses at wholesale – that is, in large quantities.
- **Retailer** – The products then are resold at retail – that is, in smaller quantities – to the final consumer.

Types of Organisation (Services)

- Services are **intangible** products.
- **Examples of services :**
 - Banking, insurance, logistics, consulting, delivery and repairs, maintenance
 - Government
 - **Purpose** : formulate laws and regulations
 - **Examples** : public safety, policing, fire protection
 - Non-Profit Organisations
 - **Purpose** : charitable by providing goods and services on voluntary basis
 - **Examples** : education, medical care, religious services, cultural activities

Business Functions of an Organisation



Business Functions - Examples

- Marketing Functions

- Determine **prices** of products and services
- **Advertising** and promotions of products and services
- Carry out **promotions**
- Carry out market **surveys** of customers' needs or change in tastes and trends
- Carry out market **intelligence** of competitors

- Sales Functions

- Answer **enquiries** from customers.
- Create **quotations** (40K unit, price per unit :RM2K) for customers.
- Accept **orders** from customers eg. create stock and customer credit status.
- **Collaborate** with other departments for delivery of orders, billing and collecting payments.

order : 001 20/11/2022 customer 1 : wholesaler 1 product 1- 50K units- RM1- RM50K			
order :002 26/11/2022 product 2- 5K units- RM1 -RM5K, Product 3 -5K units -RM1-RM5K			
		First Day	Today 29/11/2022
customer 1 credit limit :100k			available credit 100k-50K-5K-5K=40K
customer 1(wholesaler)			
paid RM 60K		30/12/2022	

Business Functions – Examples

- **Production Functions**
 - **Production planning** – deciding what (products) to produce, how much to produce and when to produce.
 - **Capacity planning** – determining capacity requirements (machines and human resources).
 - Determine **materials** requirements.
 - **Monitoring** and control production progress.
- **Human Resource Functions**
 - Assessment of staff **requirement**
 - **Recruitment** of staff eg. evaluation and selection
 - **Training and development** eg. create training plan, carry out training, record training, conduct training needs analysis
 - **Payroll** eg calculate gross and net pay, create payslip, compute deductions.
 - **Staff performance appraisal**

	Order :001		5/12/2022	
Hong Kong	customer 1 (wholesaler)		PC model 1 - 2000 unit	
Transfer order	KL HQ -warehouse	JH branch		
PC model1 - available stock quantity	1800 unit	200 unit		
Transfer order	KL HQ	JH	KL factory /plant	KL factory /plant
			check the available material : enough	check the available material :not enough
PC model1 - available stock quantity	1800 unit	0	harddisk :300units	harddisk : 50units
			DVD drive :300 units	DVD drive :100 units
			network card : 400 units	network card : 100 units
			make 200 PC : 1 week	make 200 PC : 1 month

- 1) Create transfer order for controlling the movement of goods within the warehouse
- 2) Information Systems allows customising the checking type

– **Production planning** – deciding what (products) to produce, how much to produce and when to produce.

Manufacturing and Production Systems

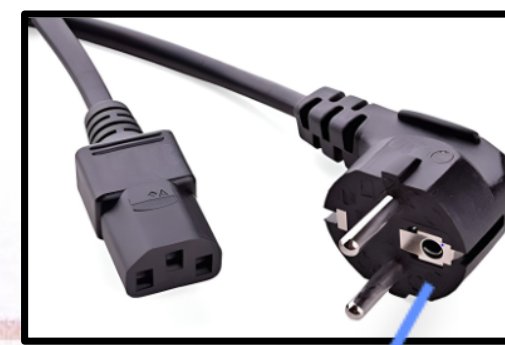
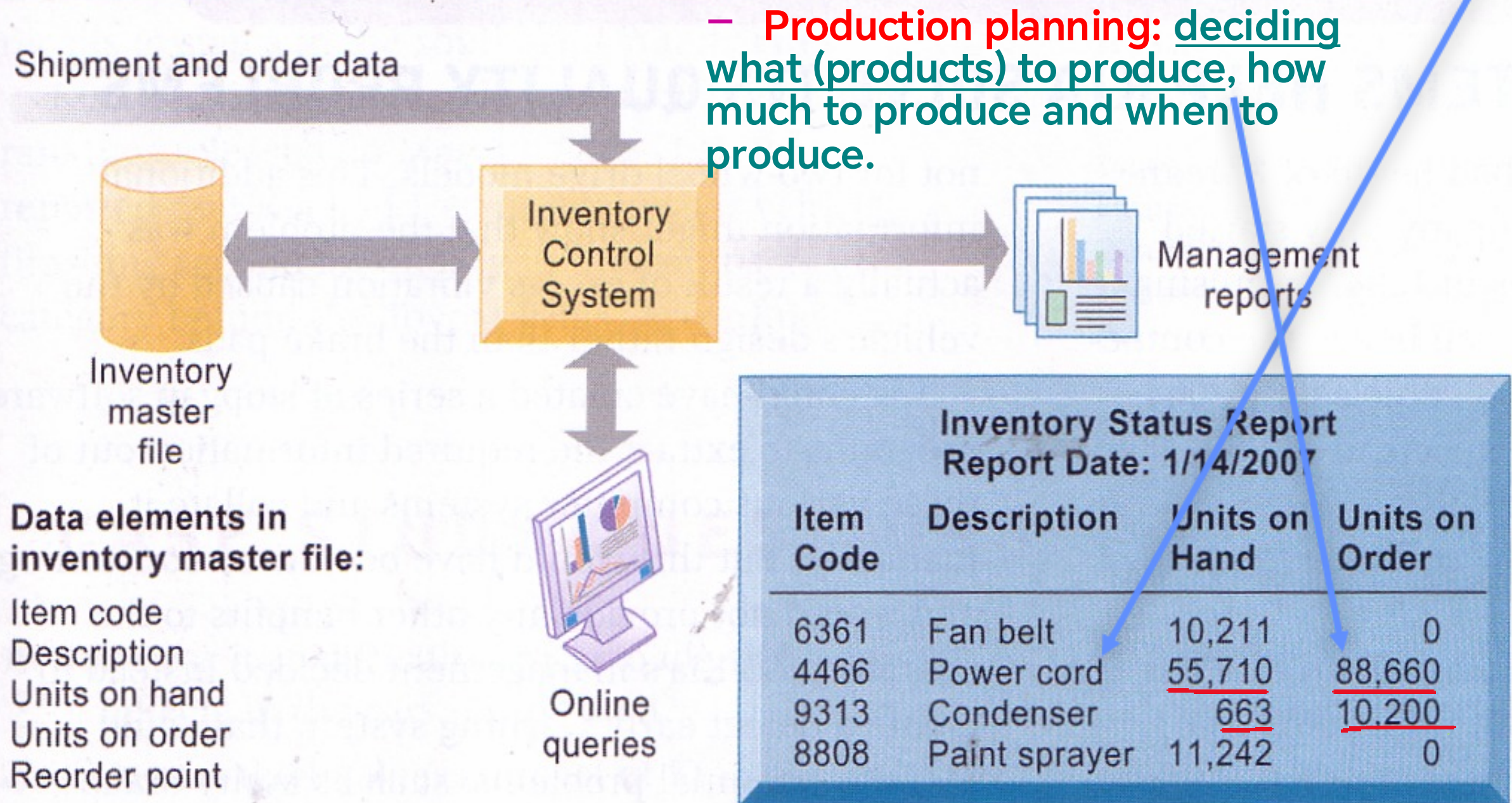


FIGURE 2-3 OVERVIEW OF AN INVENTORY SYSTEM



This system provides information about the number of items available in inventory to support manufacturing and production activities.

Business Functions - Examples

- Accounting Functions

- **Recording** business transaction entries (debits and credits)
- **Maintain** assets records (items the business owns eg. cash, equipment, and buildings)
- **Maintain** liabilities records (eg. items business owe others)
- **Reporting** – financial profitability (eg P&L) and financial *state* of business (eg. balance sheets)

inventory management system						accounting system	
	inventory system	stock count (actual qty)	difference	Price	Cost	Stock value	Lost
PC model 1	2000 units	1900 units	2000-1900=100 units	RM1000	RM500	1900 units *1000 = RM190 K	100*1000 =RM100 K

Centerfield Sporting Goods
General ledger - #1000 Cash account
(Partial - January 2020)

Date	Journal Entry #	Description	Debit	Credit	Balance
1-Jan	N/A	Beginning cash balance			80,000
1-Jan	#1	Purchase inventory		10,000	70,000
5-Jan	#2	Payment on accounts payable		4,000	66,000
9-Jan	#3	Cash received from customer	6,000		72,000

(Note: Ending cash balance is posted on 1/31, after all January activity is posted)

Business Functions – Examples

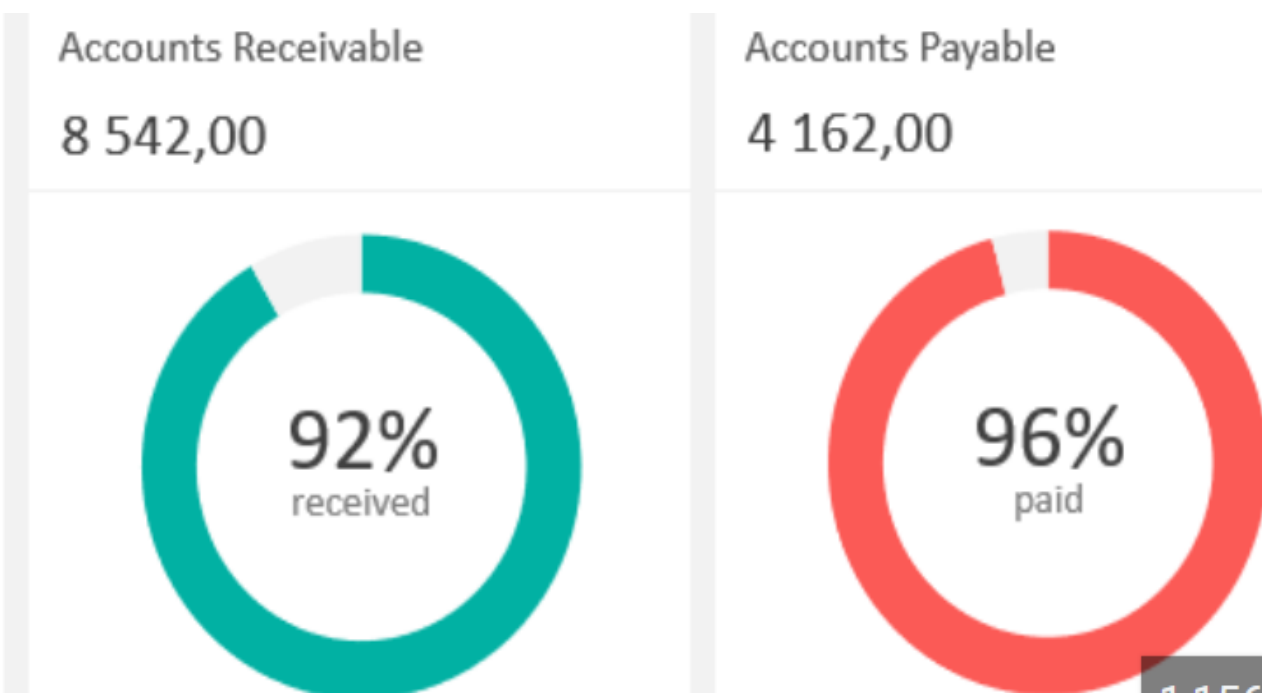
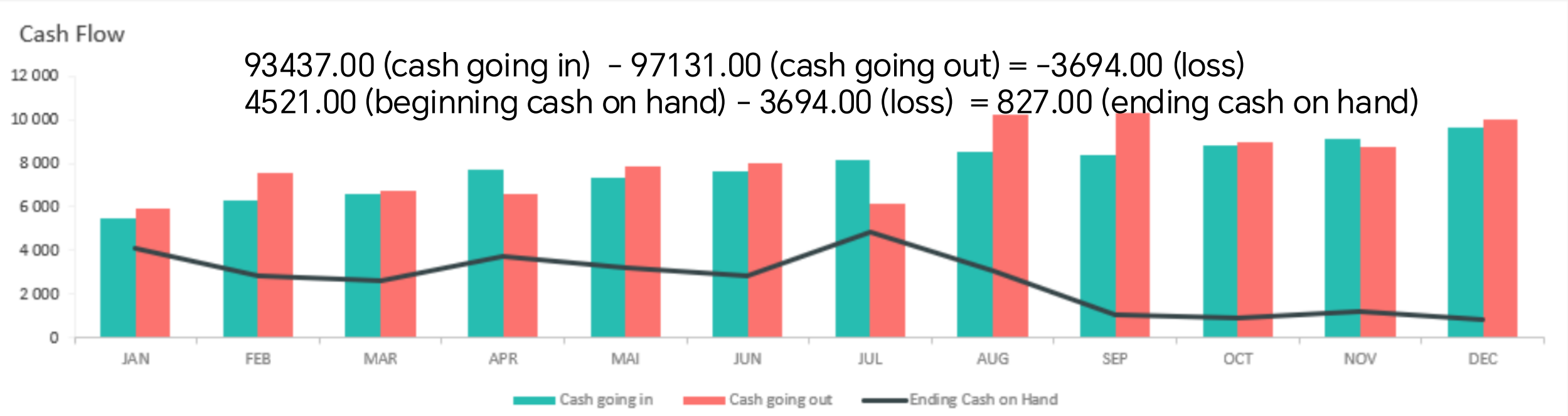
- Finance Functions
 - Carry out budgeting
 - Obtaining money needed by the business
 - Sale of goods and services eg. profits earned
 - Investments made by the business eg shares
 - Banks and other institutions eg. loans
 - Plan and control the use of money (cash flows).

3. Cash Flow Dashboard - YTD

Last month with data: December

Cash Flow Summary report

Beginning Cash on Hand	Cash going in	Cash going out	Profit / Loss	Ending Cash on Hand
4 521,00	93 437,00	97 131,00	-3 694,00	827,00



Payroll program/software computes payroll and generates payslip for employee.



ABC SDN. BHD.

52, Jalan Awan Hijau, 58200 Kuala Lumpur, Malaysia

03-03-2017

NAME : ROWAN SEBASTIAN ATKINSON

STAFF NO : 058

NRIC No : 800106 - 10 - 5821

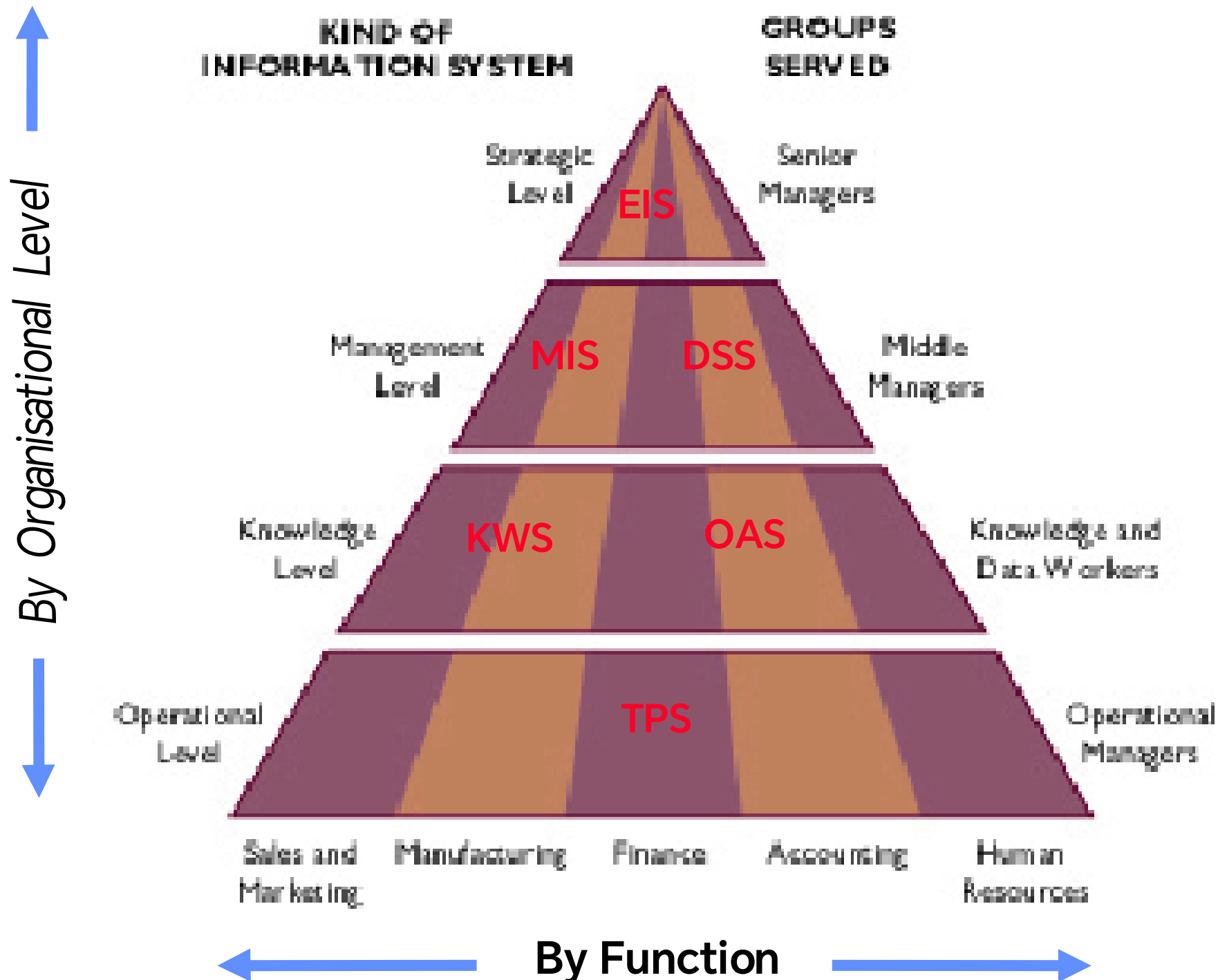
Payslip for February, 2017

EARNINGS	CURRENT (RM)	DEDUCTIONS	CURRENT (RM)
Basic Pay	2,500.00	Employee EPF Employee SOCSO	275.00 12.25
Gross Total	2,500.00		
Employer EPF Employer SOCSO	325.00 42.85	Total Deductions	287.25
Medical Leave = 1, Balance = 9 Annual Leave = 3, Balance = 17		NETT PAY	2,212.75

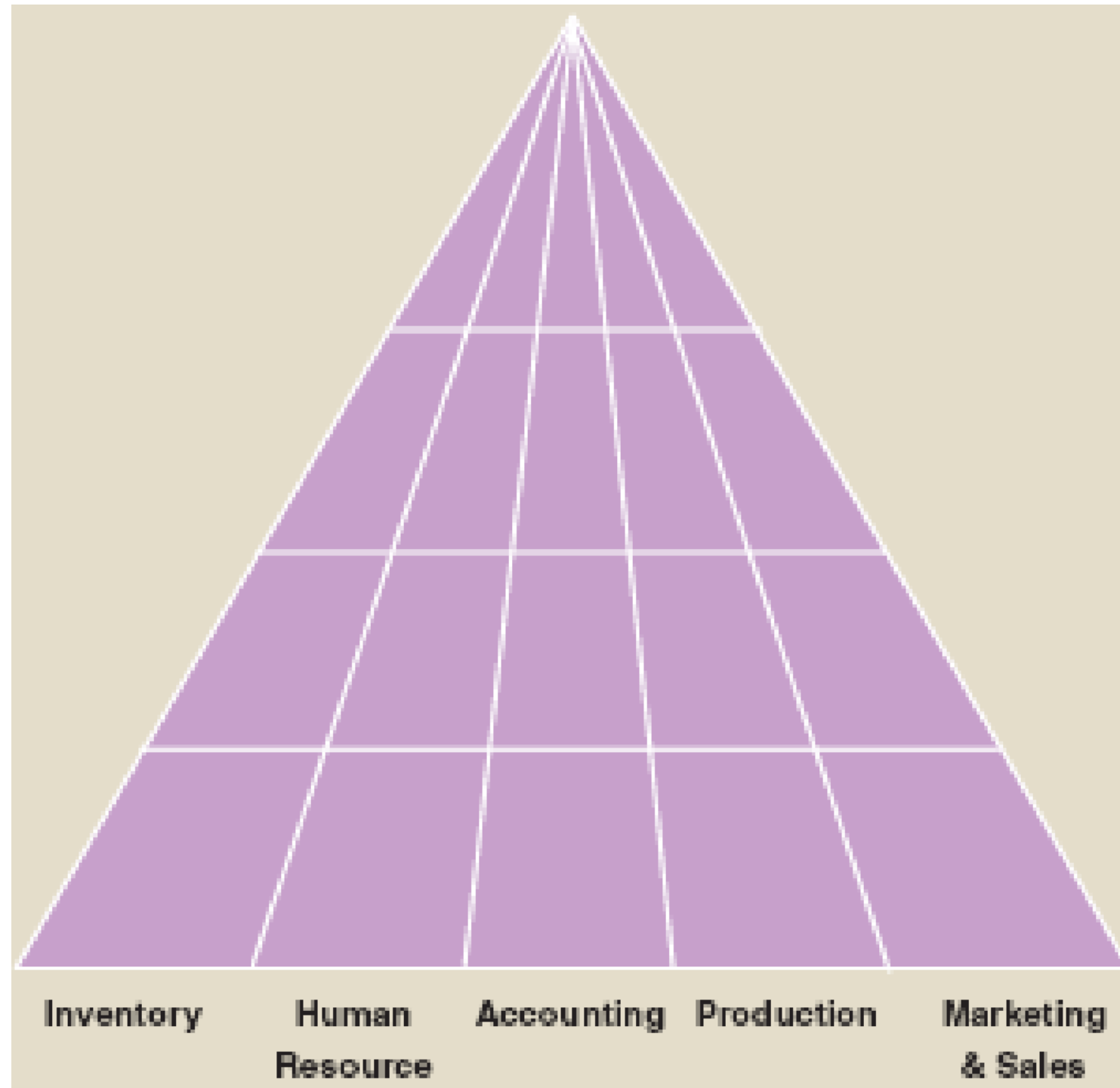
Business Functions – Examples

- **Information Technology Functions**
 - Analysis and Design of users' needs
 - Developing software
 - Buy packages/software
 - Implementing eg. training, data migration
 - Operation – assisting other functional areas (depts) on using IT.
 - Software/hardware maintenance

Information Systems Used (categories)



ISs by Functional Areas



Examples of ISs by Functions

- Purchasing System (buy)
- Sales System (sell)
- Inventory System (store)
- Production System (make)
- Accounting System (track)
- Marketing System (promote)
- HR System (manage)

Exercise for Students

- List any 4 functional systems for an **airline**
- List any 4 functional systems for a **manufacturing** company
- From the list indicate which systems are 'core' and which are 'non-core' .

Solutions

- List any 4 functional systems for an airline
 - Air-ticket reservation system (core)
 - Check-in system (core)
 - Flight scheduling system (core)
 - Plane maintenance system (core)
 - Accounting system (non-core)
 - HR system (non-core)
- List any 4 functional systems for a manufacturing company
 - Production system (core)
 - Sales system (core)
 - Purchasing system (core)
 - Inventory system (core)
 - CAD system (core)
 - HR system(non-core)

Enterprise Systems (Characteristics)

- **System.** Single large system consisting of many modules eg. ERP, CRM, SCM systems.
- **Integrate.** ES integrates business *processes*, *databases*, and *functional areas* (internally and externally). Allows data flows (vertically and horizontally).
- **Data Storage.** ES provides a *centralized database storage* for all users (intra or inter-companies).
- **Data Sharing.** ES allows users to share information *seamlessly* no matter where the data is located and used for decision making eg. inventory data shared with sales and purchasing staff.
- **Purpose.** ES *automates* business *processes* of the entire enterprise eg. order-to-cash process.

Types of Enterprise Systems

- Enterprise Resource Planning (**ERP**) system
- Customer Relationship Management (**CRM**) System
- Supply Chain Management (**SCM**) system

What is ERP ?

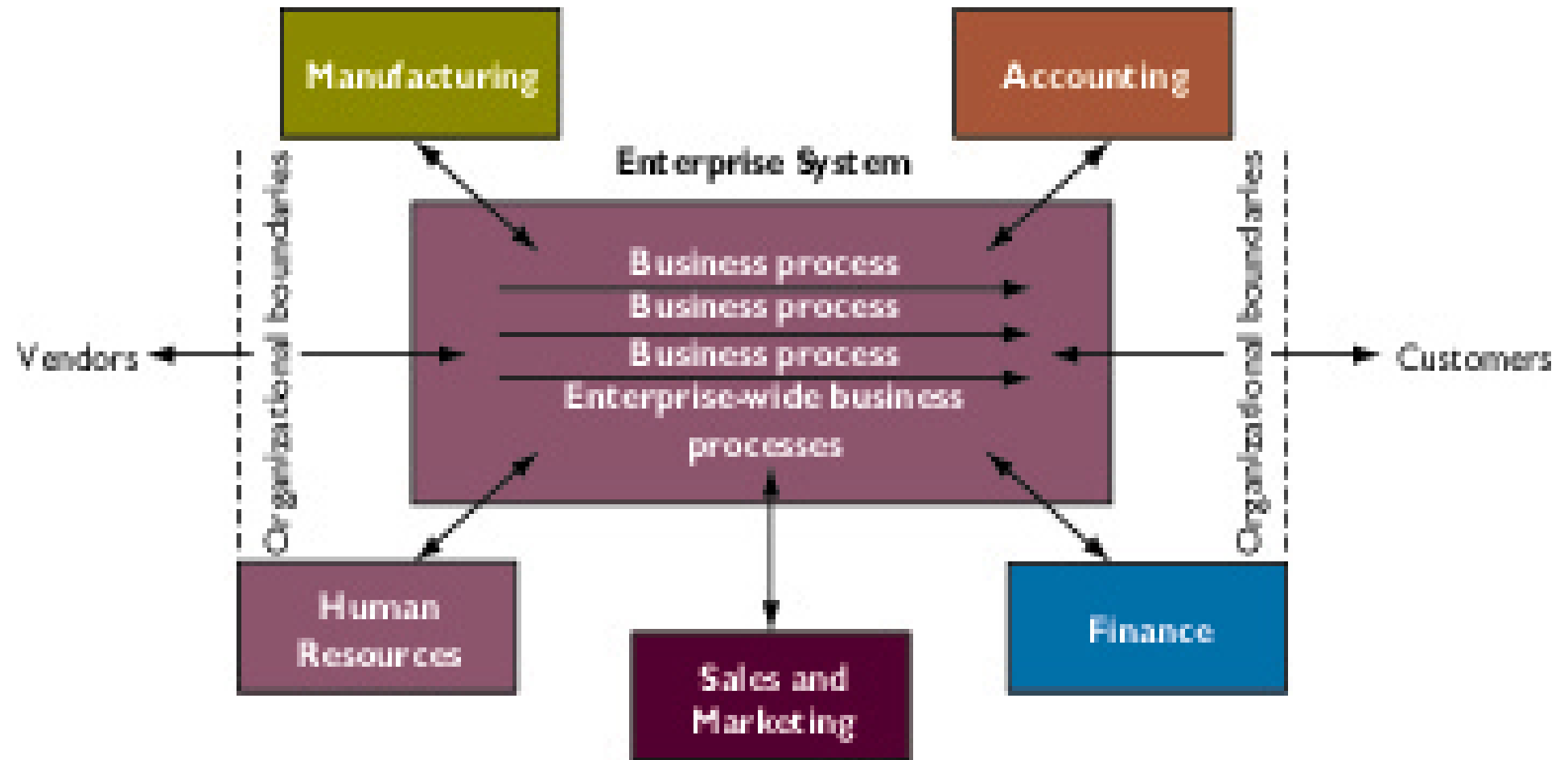
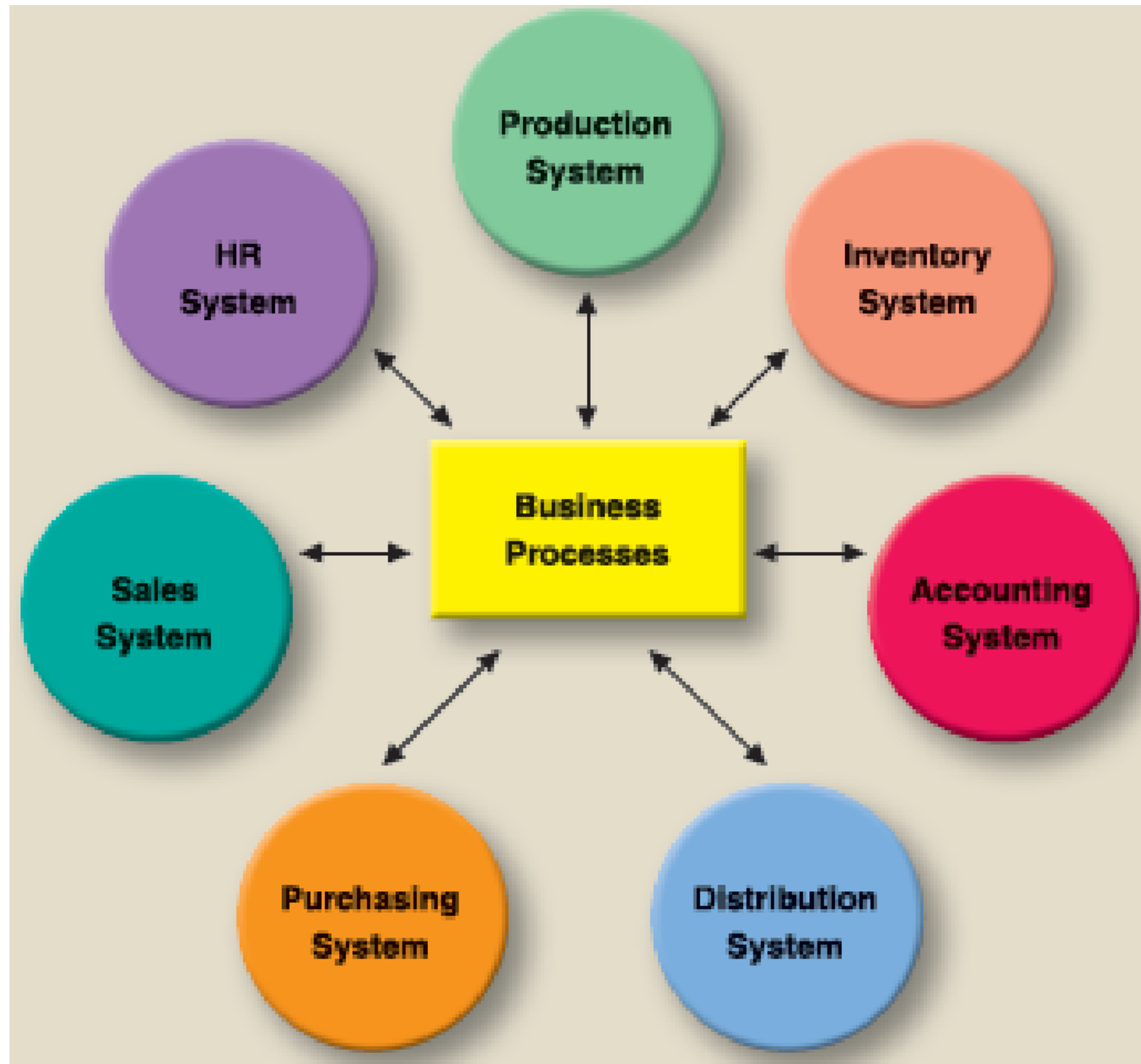


Diagram: Adapted from Laudon and Laudon

ERP **integrate** the key business processes of an entire firm into a single large software system that allows information to **flow** seamlessly throughout the organisation. These systems may **include** (connect) transactions with customers and vendors

Integrated Processes in ERP

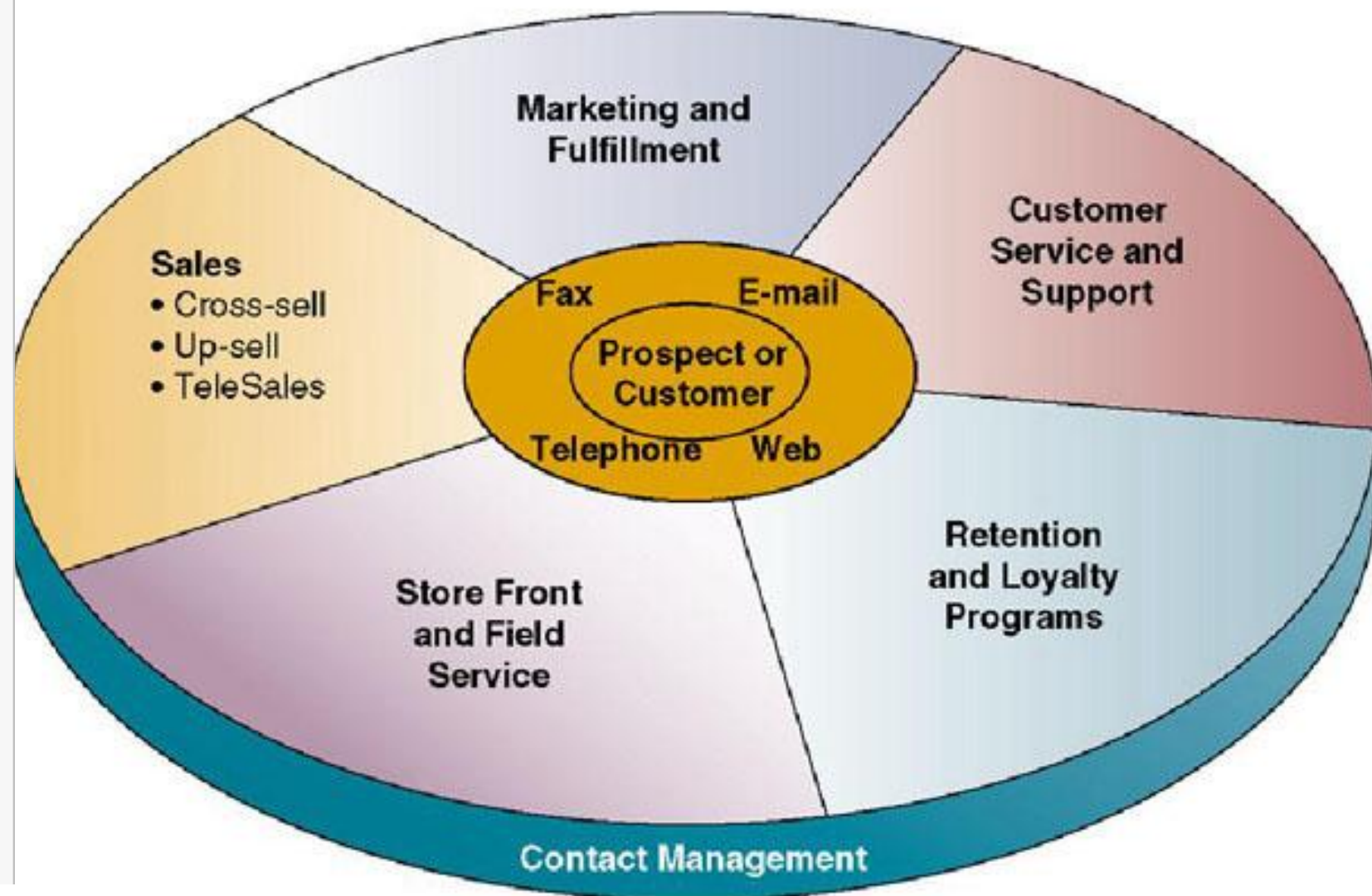


What is CRM ?

- Uses **technology** (data mining) to look (analyse) at customers from a multifaceted perspectives eg. buy patterns, habits, income.
- Uses a set of integrated **applications** to address all issues about customers *relationship* in: sales, marketing and service.

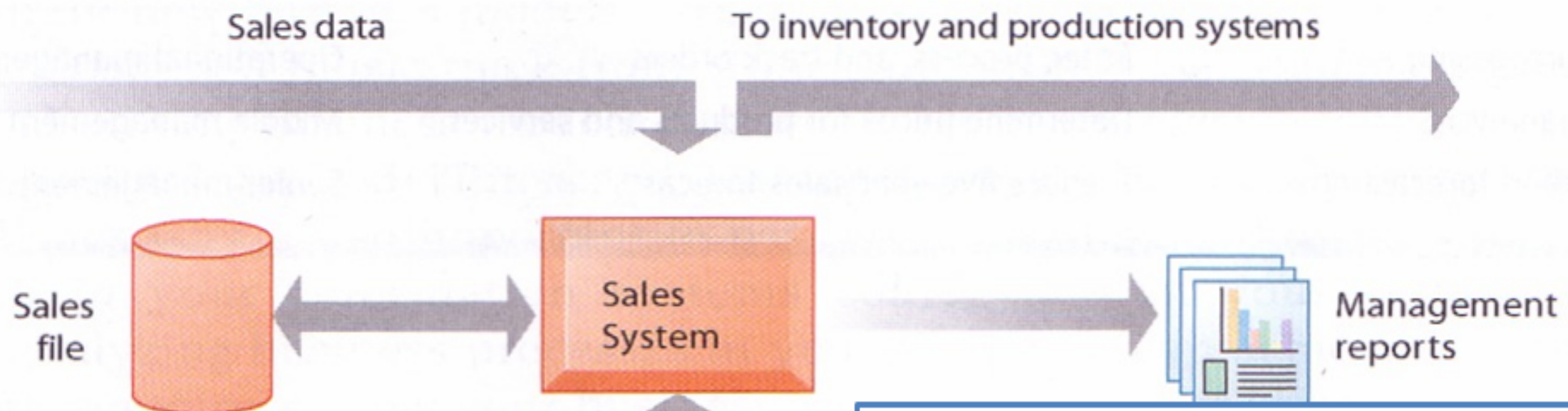
Typical Modules of CRM

- CRM systems consists of a family of software modules that perform business activities involved mainly in **front office** processes.



Typical Modules of CRM

- **Marketing.** CRM automates various tasks related to *pre-selling* activities eg. qualifying leads, scheduling sales contacts with customers, and providing information.
- **Sales.** CRM helps to answer enquiries, create quotations and accept orders from customers. Carry out potential up-selling and cross-selling tasks.
- **Service.** CRM helps in *post-selling* activities to provide good customer service eg. creation of service requests, assign qualified personnel.



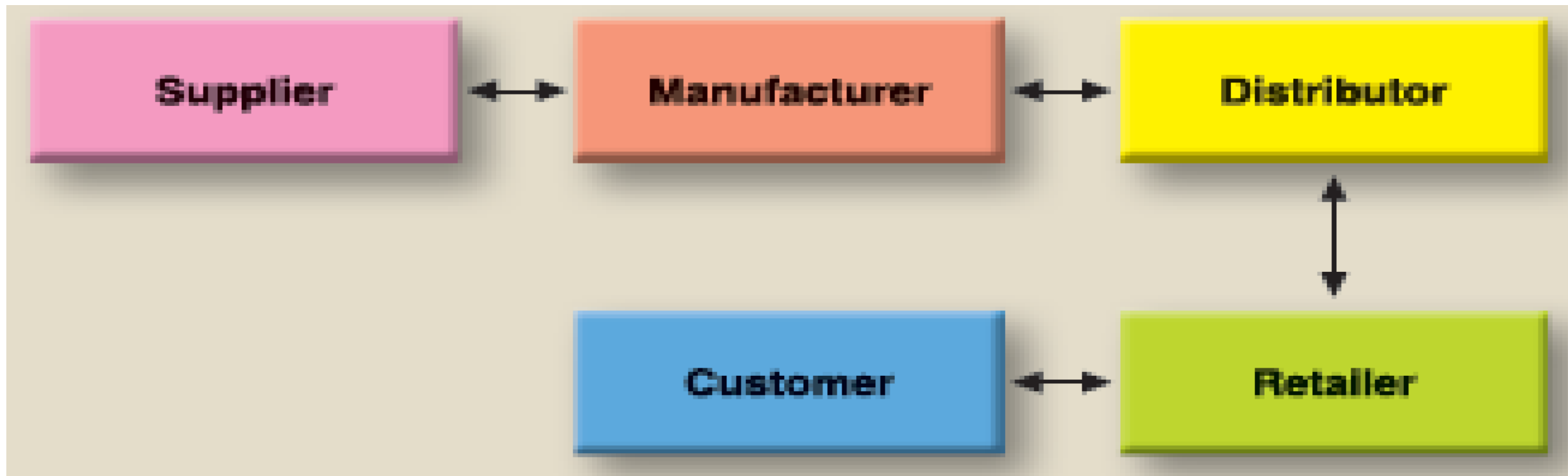
Data elements in sales file:

Store number
Item number
Item description
Color
Size
Unit price
Units sold
Date

This system captures sales data at the moment the sale takes place to help the business monitor sales transactions and to provide information to help management analyze sales trends and the effectiveness of marketing campaigns.

Sales Report: Week Ending 2/7/07						
Item No.	Description	Color	Size	Unit Price	Units Sold	Total Sales
294	Sports bag	Black	Small	10.00	10,451	\$104,510
295	Sports bag	Black	Medium	20.00	21,800	\$436,000
394	Sports bag	Red	Small	10.00	5,331	\$53,310

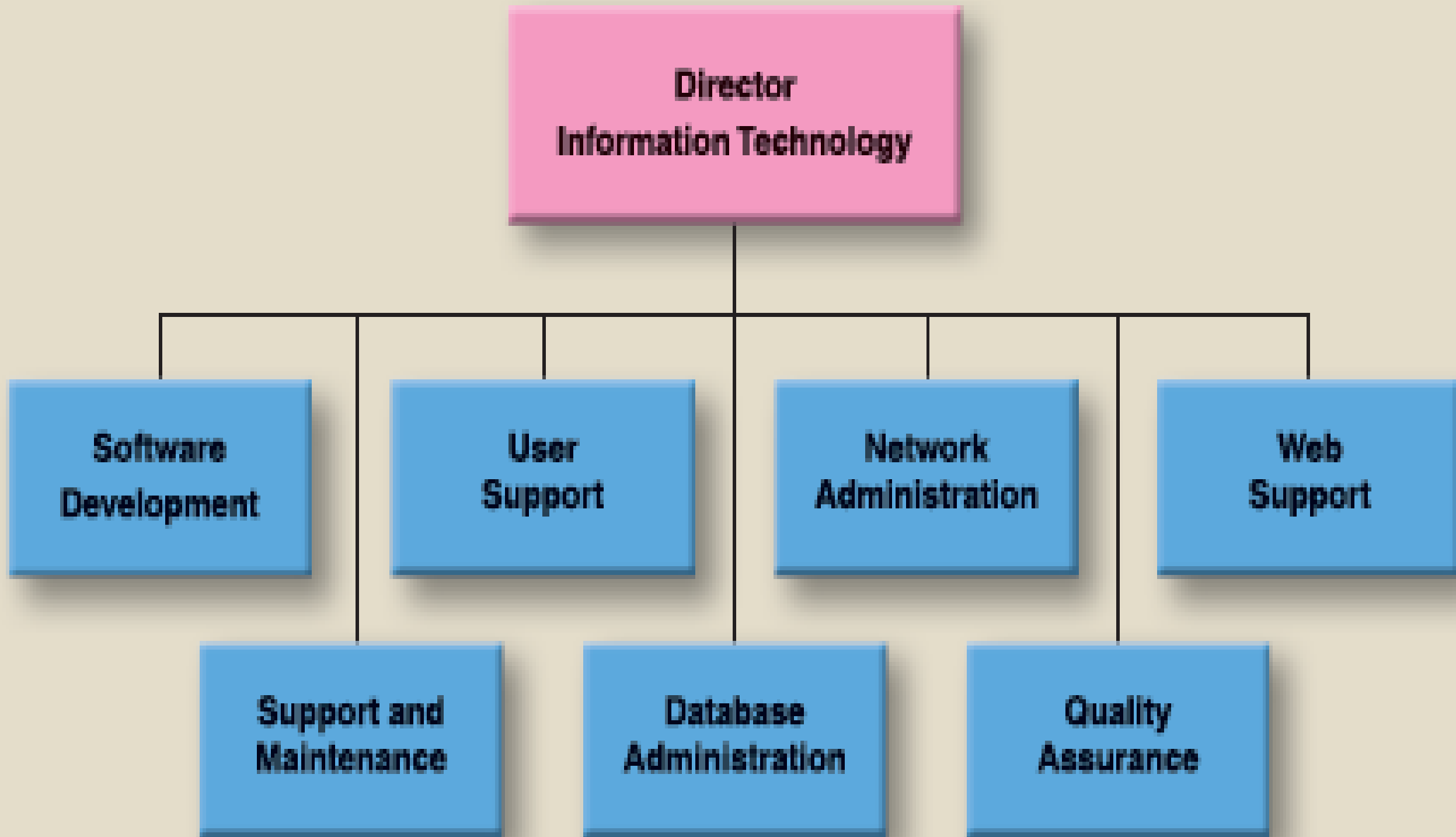
Supply Chain Management (SCM)



- **Purpose** – to deliver product more rapidly to customers, to reduce time, redundant effort, inventory costs
- **Linkage of activities** : buying, making, moving product
- **Integrates** – supplier, manufacturer distributor, retailer and customer
- **Network of facilities** – procuring raw materials, transforming and distributing products to customers.

IT Department

IT Department (Functional Areas)

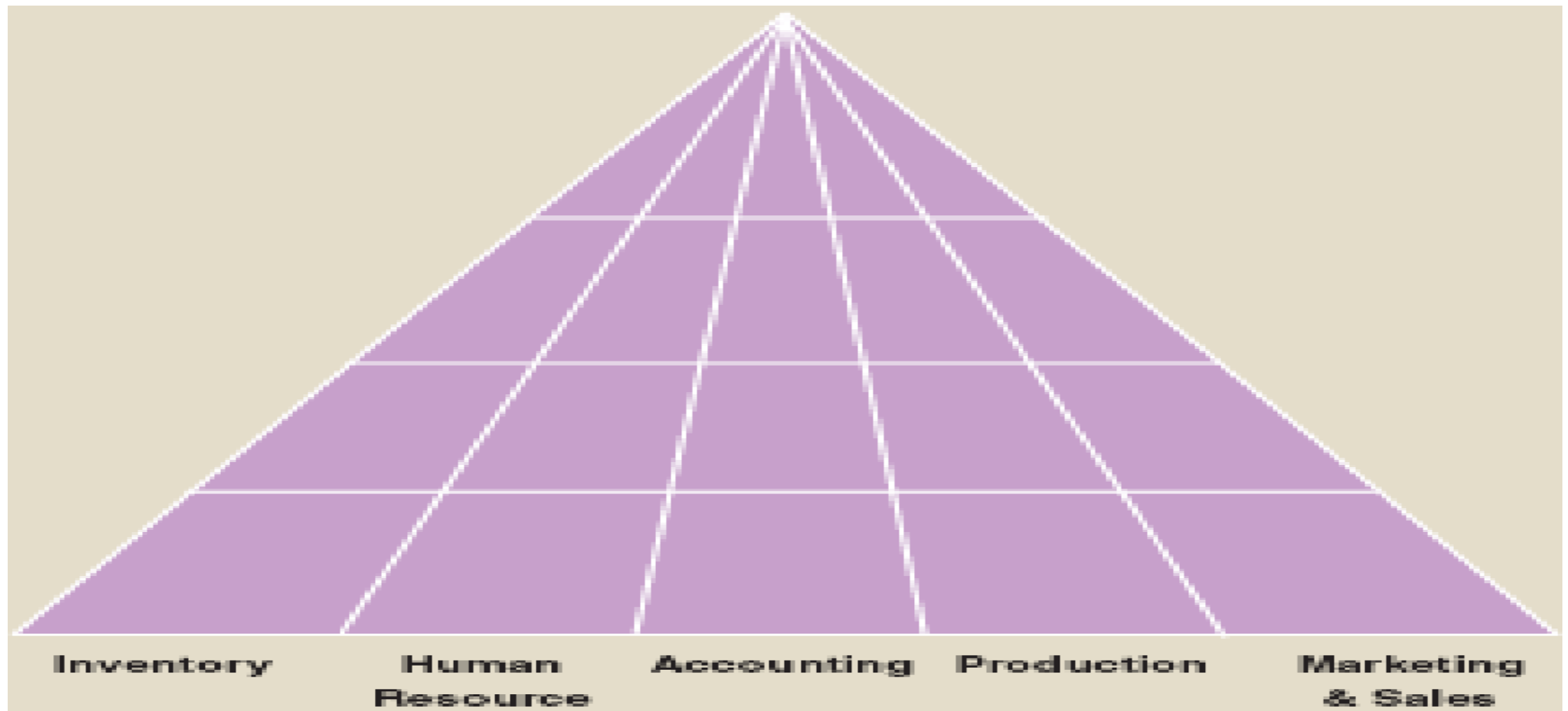


IT Department (Functional Areas)

1. **Software Development** – teams use tools to develop programs
2. **Quality Assurance (QA)** – ensure quality of software developed, monitor changes, control quality standards
3. **Web Support** – support Intranets, Extranet operations, EC, develop websites
4. **Database Administration** – database design, backup, access and security control
5. **Network Administration** – maintenance and support eg. control user access, configuring, monitoring
6. **Operation Support** – eg training, operating, help desks, trouble shooting.
7. **Maintenance** – for both hardware and software eg. PCs, networks, ISs

Organisation of a Business

- **Organisation Structure** – *employees* are grouped based on departmental **functions** eg. Accounting, Sales etc.
- **Organisation Levels** – *employees positions* are grouped into vertical levels for **authority** control eg. HQ, subsidiaries (a company that is owned by a larger company), **branches, departments, product lines, regions, workgroups** etc.



- *Central Region*
 - Kuala Lumpur (federal territory)
 - Putrajaya (federal territory)
 - Selangor
- *Northern Region*
 - Kedah
 - Penang
 - Perak
 - Perlis
- *Southern Region*
 - Johor
 - Melaka
 - Negeri Sembilan

Example of Organisation Chart



SDLC

Brief Introduction to SDLC

Phase 1

System Request

System Planning

Phase 2

System Analysis

Preliminary Investigation Report

Phase 3

System Design

System Requirements Specification

Phase 4

System Development

System Design Specification

Phase 5

System Implementation

Completed System

Phase 6

Operation and Maintenance

Operational System

